

Code:

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class HospitalAppointment extends Frame implements ActionListener {
```

```
    Label l1, l2, l3, l4, l5, l6, l7;
```

```
    TextField t1, t2, t3, t4;
```

```
    Checkbox male, female;
```

```
    CheckboxGroup gender;
```

```
    Choice department, doctor;
```

```
    Button book, clear;
```

```
    TextArea result;
```

```
    HospitalAppointment() {
```

```
        setTitle("Hospital Appointment Management System");
```

```
        setSize(500, 600);
```

```
        setLayout(new GridLayout(9, 2, 10, 10));
```

```
        // Labels
```

```
        l1 = new Label("Patient ID:");
```

```
        l2 = new Label("Patient Name:");
```

```
        l3 = new Label("Age:");
```

```
        l4 = new Label("Gender:");
```

```
        l5 = new Label("Department:");
```

```
l6 = new Label("Doctor Name:");
```

```
l7 = new Label("Appointment Date:");
```

```
// TextFields
```

```
t1 = new TextField();
```

```
t2 = new TextField();
```

```
t3 = new TextField();
```

```
t4 = new TextField();
```

```
// Gender
```

```
gender = new CheckboxGroup();
```

```
male = new Checkbox("Male", gender, false);
```

```
female = new Checkbox("Female", gender, false);
```

```
Panel genderPanel = new Panel();
```

```
genderPanel.add(male);
```

```
genderPanel.add(female);
```

```
// Department
```

```
department = new Choice();
```

```
department.add("Cardiology");
```

```
department.add("Neurology");
```

```
department.add("Orthopedics");
```

```
department.add("General Medicine");
```

```
department.add("Pediatrics");
```

```
// Doctor
```

```
doctor = new Choice();
```

```
doctor.add("Dr. Kumar");
```

```
doctor.add("Dr. Meena");
```

```
doctor.add("Dr. Raj");
```

```
doctor.add("Dr. Priya");
```

```
// Buttons
```

```
book = new Button("Book Appointment");
```

```
clear = new Button("Clear");
```

```
book.addActionListener(this);
```

```
clear.addActionListener(this);
```

```
// Result area
```

```
result = new TextArea();
```

```
result.setEditable(false);
```

```
// Add components
```

```
add(l1);
```

```
add(t1);
```

```
add(l2);
```

```
add(t2);
```

```
add(l3);
```

```
add(t3);
```

```
add(l4);
```

```
add(genderPanel);
```

```
add(l5);
```

```
add(department);
```

```
add(l6);
```

```
add(doctor);
```

```
add(l7);
```

```
add(t4);
```

```
add(book);
```

```
add(clear);
```

```
add(new Label("Appointment Summary:"));
```

```
add(result);
```

```
// Window closing
```

```
addWindowListener(new WindowAdapter() {
```

```
    public void windowClosing(WindowEvent e) {
```

```
        dispose();
```

```
    }
```

```
});
```

```
setVisible(true);  
}
```

```
public void actionPerformed(ActionEvent e) {
```

```
    // Book Appointment
```

```
    if (e.getSource() == book) {
```

```
        String patientId = t1.getText().trim();
```

```
        String patientName = t2.getText().trim();
```

```
        String ageText = t3.getText().trim();
```

```
        String appointmentDate = t4.getText().trim();
```

```
        // Validate Patient ID
```

```
        if (patientId.equals("")) {
```

```
            result.setText("Patient ID Required");
```

```
            return;
```

```
        }
```

```
        // Validate Patient Name
```

```
        if (patientName.equals("")) {
```

```
            result.setText("Patient Name Required");
```

```
            return;
```

```
        }
```

```
// Validate Age

if (ageText.equals("")) {
    result.setText("Age Required");
    return;
}

int age;

try {
    age = Integer.parseInt(ageText);
} catch (NumberFormatException ex) {
    result.setText("Invalid Age");
    return;
}

if (age <= 0) {
    result.setText("Invalid Age");
    return;
}

// Validate Gender

if (gender.getSelectedCheckbox() == null) {
    result.setText("Gender Required");
    return;
}
```

```
// Validate Appointment Date

if (appointmentDate.equals("")) {

    result.setText("Appointment Date Required");

    return;

}
```

```
String selectedGender =

    gender.getSelectedCheckbox().getLabel();
```

```
String selectedDepartment =

    department.getSelectedItem();
```

```
String selectedDoctor =

    doctor.getSelectedItem();
```

```
// Appointment Summary

result.setText(

    "Appointment Booked Successfully\n\n" +

    "Patient ID: " + patientId + "\n" +

    "Patient Name: " + patientName + "\n" +

    "Age: " + age + "\n" +

    "Gender: " + selectedGender + "\n" +

    "Department: " + selectedDepartment + "\n" +

    "Doctor: " + selectedDoctor + "\n" +

    "Appointment Date: " + appointmentDate

);
```

```
}
```

```
// Clear Button
```

```
if (e.getSource() == clear) {
```

```
    t1.setText("");
```

```
    t2.setText("");
```

```
    t3.setText("");
```

```
    t4.setText("");
```

```
    gender.setSelectedCheckbox(null);
```

```
    department.select(0);
```

```
    doctor.select(0);
```

```
    result.setText("");
```

```
}
```

```
}
```

```
public static void main(String[] args) {
```

```
    new HospitalAppointment();
```

```
}
```

```
}
```